

MySQL Enterprise Edition DBA Roadmap

1. MySQL fundamentals

First understand:

MySQL architecture

MySQL Server vs MySQL client

Storage engines, especially InnoDB

Databases, tables, indexes

SQL basics: SELECT, INSERT, UPDATE, DELETE

Users, privileges, roles

Configuration files (my.cnf / my.ini)

MySQL data directory

Ports and networking

Server startup/shutdown

Goal: Install MySQL and be comfortable managing a single server.

2. Installation & configuration

Learn to install and configure MySQL Enterprise Edition on Linux.

Important DBA topics:

Installation

↓

my.cnf

↓

mysqld

↓

Data directory

↓

InnoDB

↓

Logs

↓

Client connections

Practice:

mysql -u root -p

mysqladmin status

systemctl status mysqld

systemctl start mysqld

systemctl stop mysqld

Then learn important variables:

SHOW VARIABLES;

SHOW VARIABLES LIKE 'innodb%';

SHOW VARIABLES LIKE 'max_connections';

SHOW VARIABLES LIKE 'log%';

3. User & security administration

This is a major DBA responsibility.

Learn:

CREATE USER

ALTER USER

DROP USER

GRANT

REVOKE

CREATE ROLE
GRANT ... TO role
SET DEFAULT ROLE
For example:
CREATE USER 'appuser'@'10.0.0.%'
IDENTIFIED BY 'StrongPassword';

GRANT SELECT, INSERT, UPDATE
ON company.*
TO 'appuser'@'10.0.0.%';
Then learn authentication, password policies, SSL/TLS, auditing, and security hardening.

4. InnoDB administration

Spend significant time here.
Understand:

Buffer pool
Redo logs
Undo logs
Doublewrite buffer
Adaptive Hash Index
Transactions
ACID
MVCC
Isolation levels
Row-level locking
Deadlocks
Tablespaces
ibdata
.ibd files

You should eventually be able to explain what happens internally when you execute:

```
UPDATE employees  
SET salary = salary * 1.10  
WHERE employee_id = 100;
```

5. Backup & recovery

This is critical DBA knowledge.

Learn:

Logical backups
Physical backups
Full backups
Incremental backups
Binary logs
Point-in-time recovery
Backup validation
Restore testing

For Enterprise Edition, pay particular attention to MySQL Enterprise Backup (MEB).

A DBA should be able to handle:

Server failure

↓

Restore backup

↓

Apply binary logs

↓

Recover to desired time



Bring application back online

6. Binary logs & replication

Next learn MySQL replication thoroughly.

Topics:

Binary logging

Source/replica architecture

GTIDs

Asynchronous replication

Semi-synchronous replication

Replication filters

Replication lag

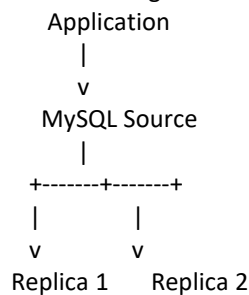
Relay logs

Failover

Switchover

Replication troubleshooting

Practice building:



7. MySQL InnoDB Cluster / high availability

After understanding basic replication, move to HA.

Learn:

MySQL Group Replication

InnoDB Cluster

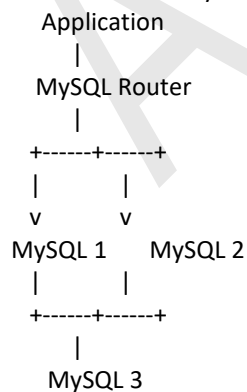
MySQL Router

Automatic failover

Read/write routing

Single-primary vs multi-primary concepts

Your lab can eventually look like:



8. Performance tuning

Then become comfortable diagnosing slow MySQL servers.

Learn:

EXPLAIN

EXPLAIN ANALYZE

Slow query log

Performance Schema
sys schema
Index optimization
Query optimization
Buffer pool sizing
Connection/thread management
Temporary tables
Disk I/O
CPU/memory issues
Lock contention
Important commands:
SHOW PROCESSLIST;

SHOW ENGINE INNODB STATUS;

EXPLAIN SELECT ...;

SELECT *
FROM performance_schema.events_statements_summary_by_digest;

9. Monitoring

Learn how a production DBA monitors MySQL.

Monitor:

CPU

RAM

Disk space

Disk I/O

Connections

Queries

QPS

Transactions

Replication lag

Locks

Deadlocks

Buffer pool

Backup status

For Enterprise Edition, learn the MySQL Enterprise Monitor ecosystem as well.

10. Enterprise-specific features

Once your core DBA skills are strong, study:

MySQL Enterprise Backup

MySQL Enterprise Monitor

MySQL Enterprise Firewall

MySQL Enterprise Audit

MySQL Enterprise Encryption

MySQL Enterprise Thread Pool

MySQL Enterprise Authentication

MySQL Enterprise Transparent Data Encryption

MySQL Enterprise Firewall

Don't start with these. Master core MySQL administration first, then learn the Enterprise features.

A good 12-week learning plan

Weeks Focus

- 1 Linux + MySQL fundamentals
- 2 Installation & configuration
- 3 Users, privileges & security

Weeks Focus

- 4 InnoDB internals
- 5 Backup & restore
- 6 Binary logs & PITR
- 7 Replication
- 8 Replication troubleshooting
- 9 Group Replication / InnoDB Cluster
- 10 Performance tuning
- 11 Monitoring & troubleshooting
- 12 Enterprise features + production scenarios

Most important point

Don't learn MySQL DBA only by reading commands. Build a 3-server lab and deliberately break things.

For example:

Linux VM 1 → MySQL Source

Linux VM 2 → MySQL Replica

Linux VM 3 → MySQL Replica

Then practice:

Create database



Create users



Configure backup



Enable binary log



Configure replication



Kill MySQL



Restore backup



Perform PITR



Create replication error



Troubleshoot



Perform failover

That will teach you much faster than memorizing commands.